ERHVERVSAKADEMI
AARHUS

Machine Learning – Day 3

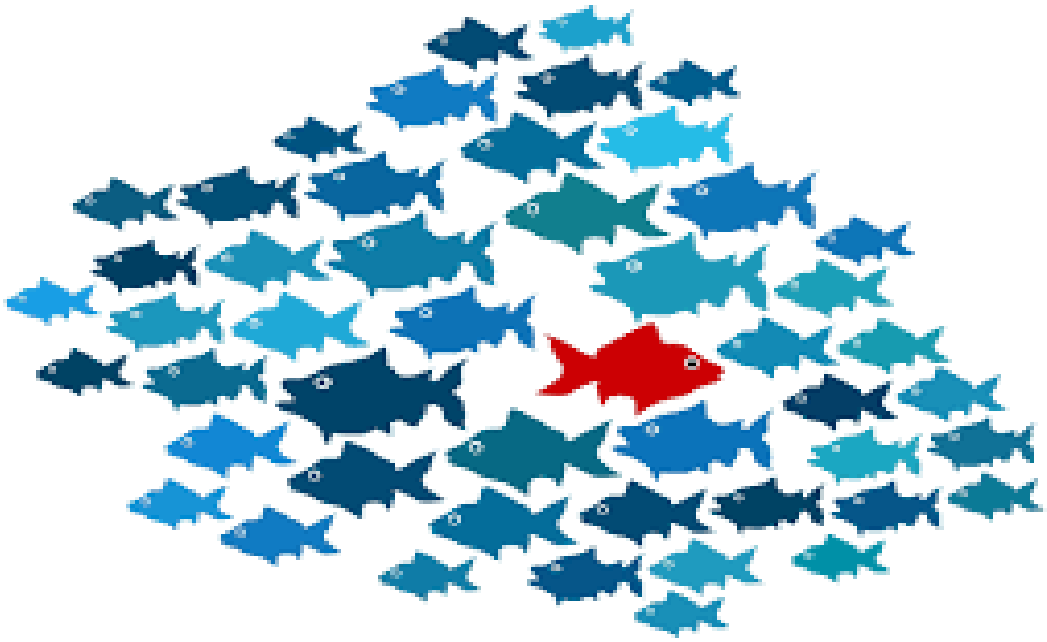
Sila, Sept 9th 2026.

Welcome - Todays Program.

- Comments about last weeks exercises.
- Chicago Crimes, real-life data.
- Introduction to Logistic Regression.
- Starting out with Logistic Regression
- Comment about SVM's
- Exercises

Recap: What can Machine Learning do?

Detect anomalies (security, fraud).



Recap: What can Machine Learning do?

Healthcare – Second opinion:

- Historical datasets of patient characteristics.
- Data and algorithm are combined to create predictive model.
- Predictive models assigns probabilities.
- Patients outcome is predicted.

Recap: What can Machine Learning do?

Customer segmentation
for targeted marketing.



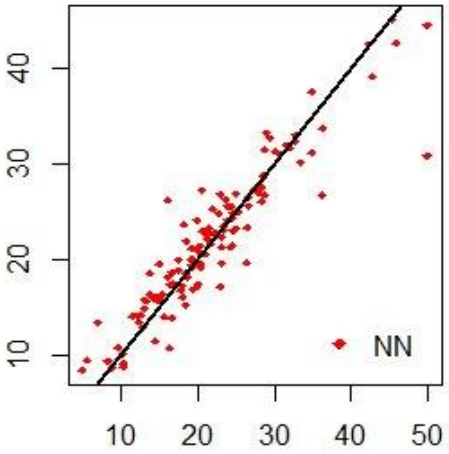
Machine Learning – How

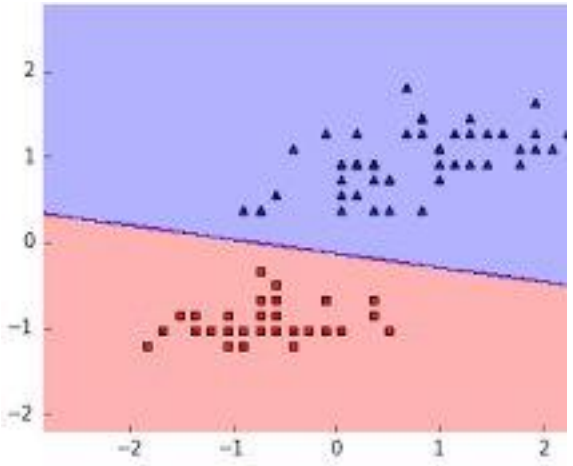
A machine learning project may not be linear, but it has a number of well known steps:

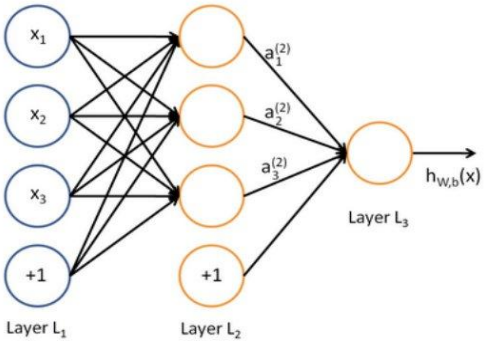
1. Define Problem.
2. Prepare Data.
3. Evaluate Algorithms.
4. Improve Results.
5. Present Results.

Machine Learning – Misc. Techniques.

Linear regression, logistic regression, neural nets and more.

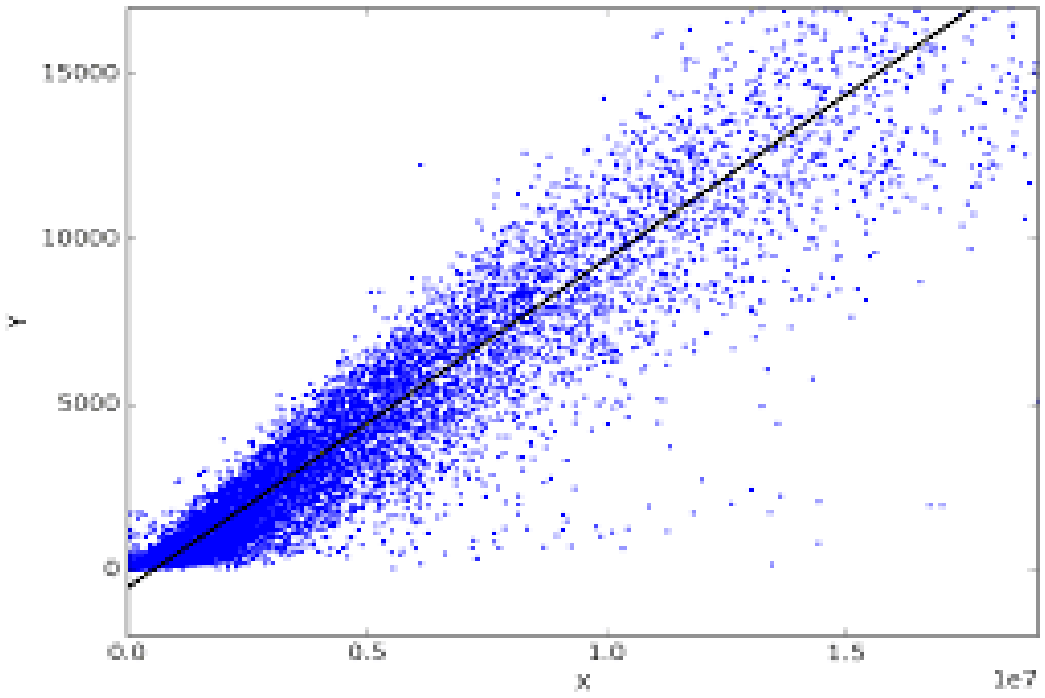






Last week: Machine Learning – Linear regression

Using linear regression:



Last week: Machine Learning – Linear regression

Gradient descent, minimize cost function.

Find the parameters for our hypothesis:

$$
\begin{aligned}
&\text{repeat until convergence } \{ \\
&\quad \theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)} \right) \\
&\quad \theta_1 := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)} \right) \cdot x^{(i)} \\
&\}
\end{aligned}
$$

Machine Learning – Linear regression

Comments to last weeks exercises?



Machine Learning – Getting to know the data.

Chicago Crime

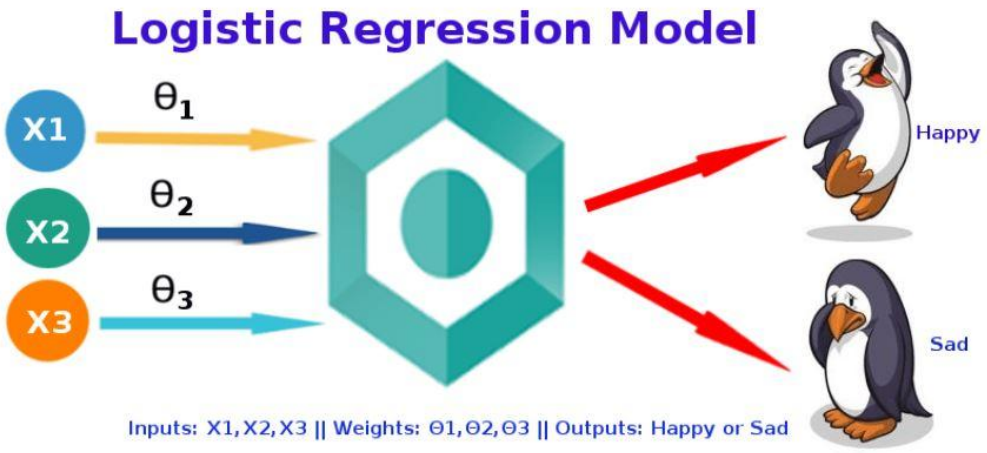


See the exercise "Chicago crimes" on Canvas.

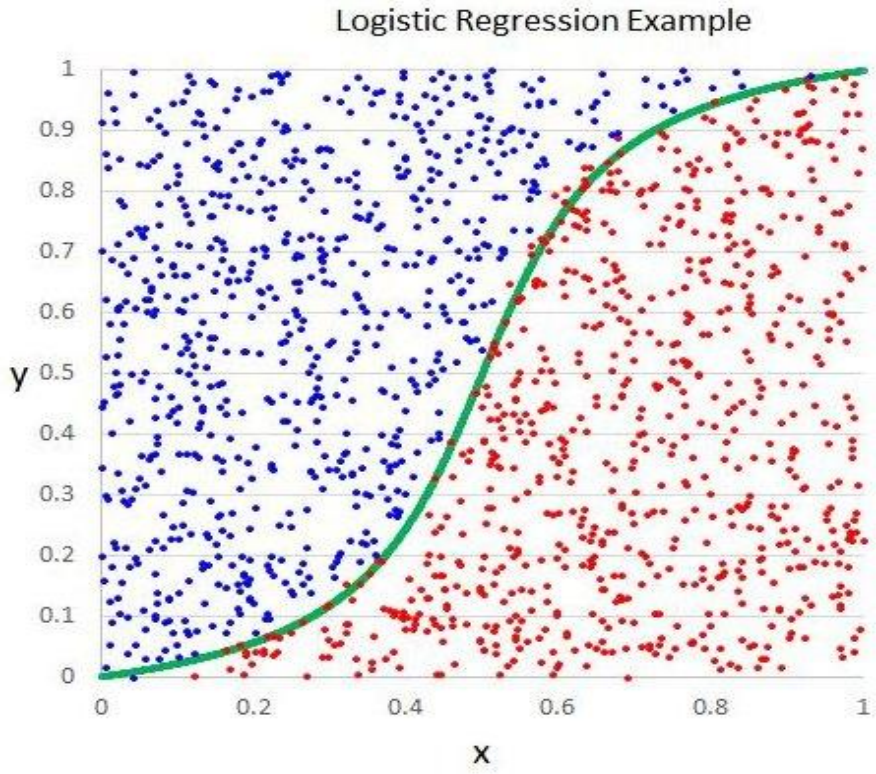
This week: Machine Learning – Logistic regression

Definition (one suggestion):

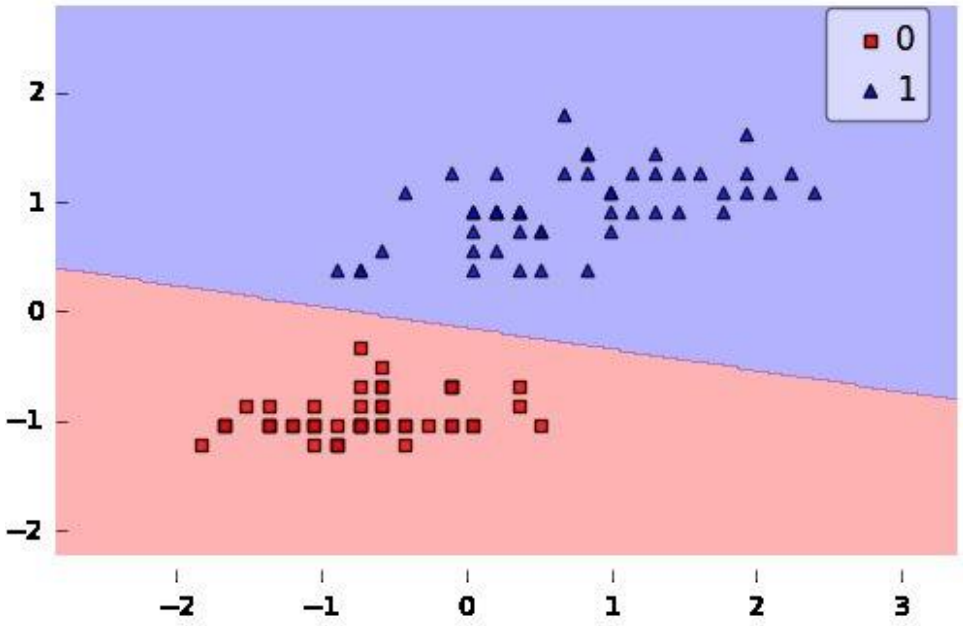
Logistic regression is used to model the probability of a certain class or event existing such as pass/fail, win/lose, alive/dead or healthy/sick.



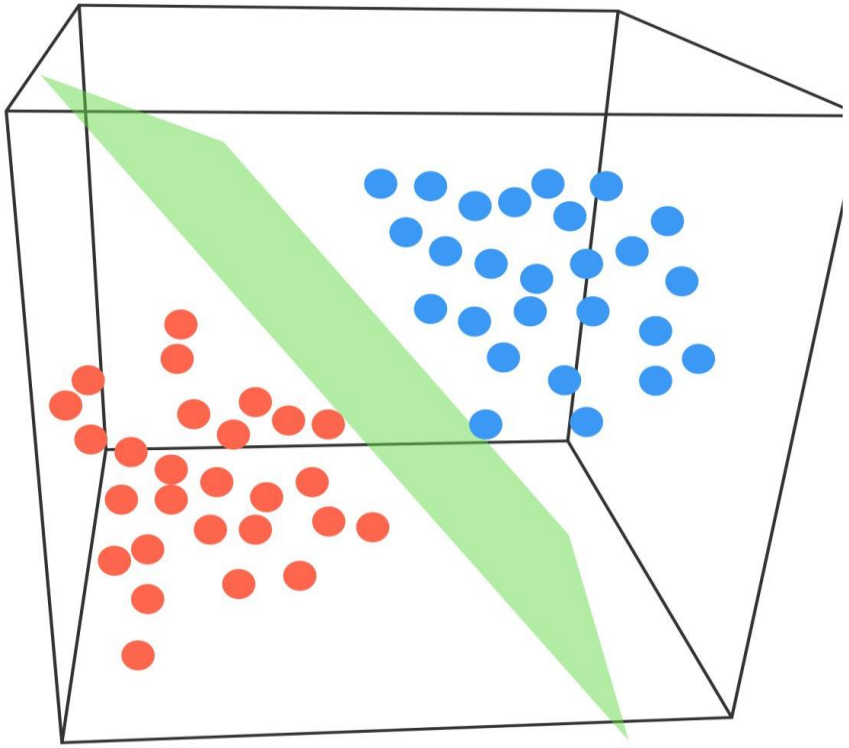
This week: Machine Learning – Logistic regression



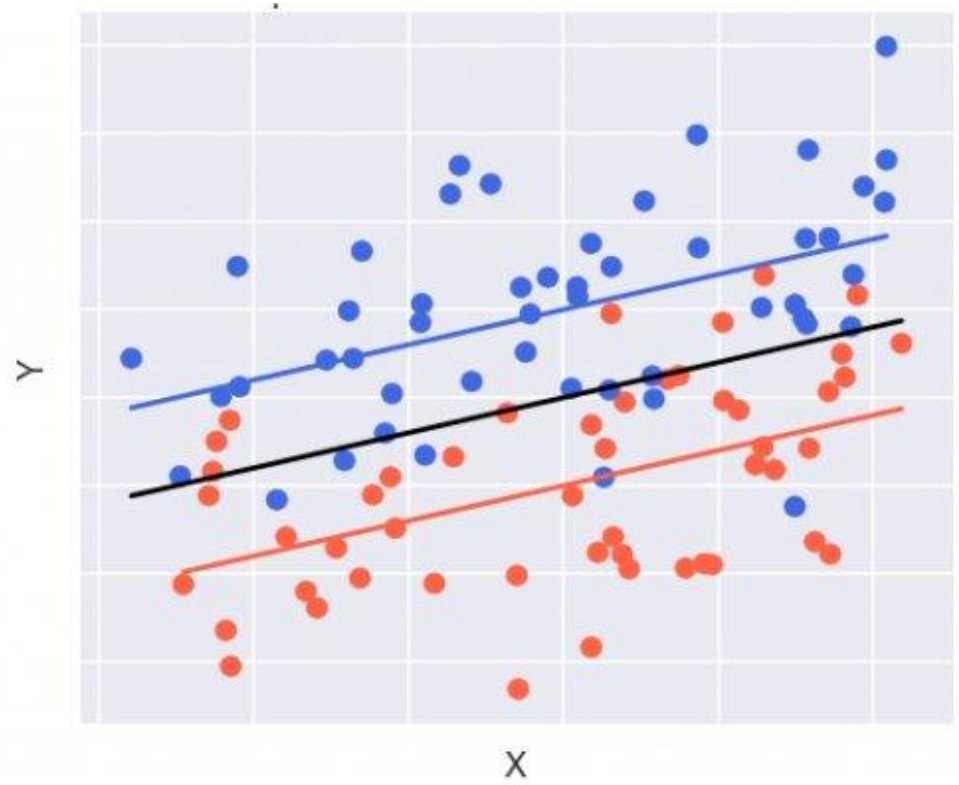
This week: Machine Learning – Logistic regression



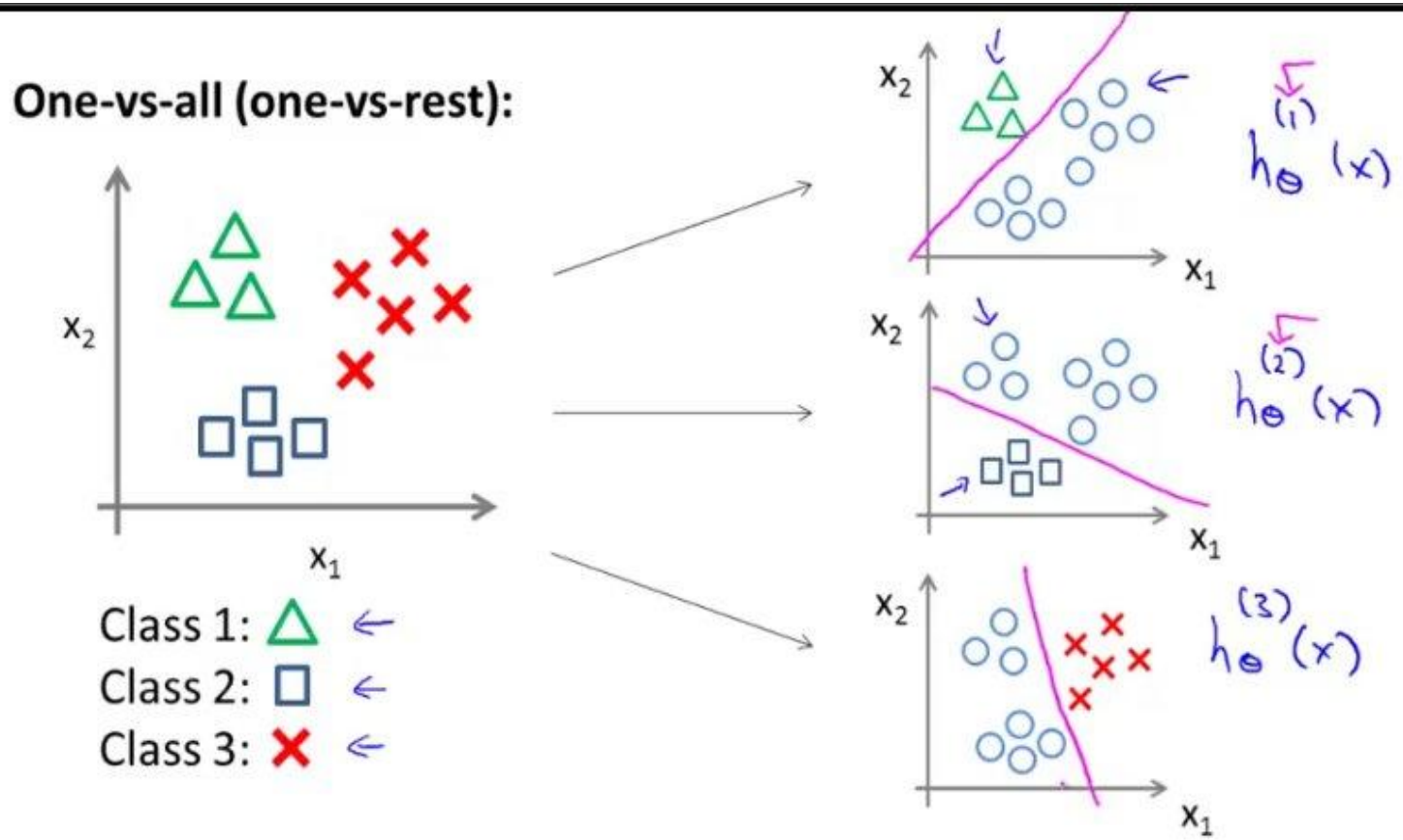
This week: Machine Learning – Logistic regression



This week: Machine Learning – Logistic regression



Machine Learning – Classifications



Machine Learning – Logistic regression

Estimating Probabilities

So how does it work? Just like a Linear Regression model, a Logistic Regression model computes a weighted sum of the input features (plus a bias term), but instead of outputting the result directly like the Linear Regression model does, it outputs the *logistic* of this result (see Equation 4-13).

Equation 4-13. Logistic Regression model estimated probability (vectorized form)

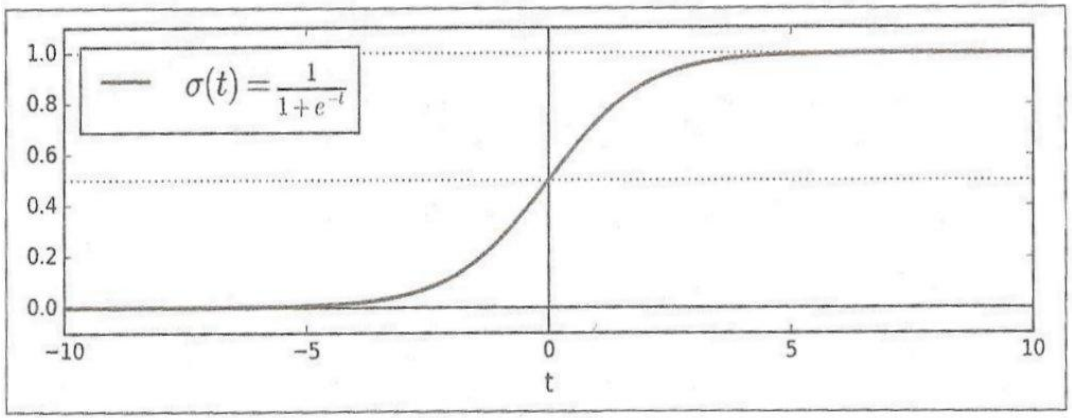
$$\hat{p} = h_\theta(\mathbf{x}) = \sigma\left(\theta^T \cdot \mathbf{x}\right)$$

The logistic—also called the *logit*, noted $\sigma(\cdot)$—is a *sigmoid function* (i.e., *S*-shaped) that outputs a number between 0 and 1.

p. 144 in "Hands on ML" by Geron

Machine Learning – Logistic regression

$$\sigma(t) = \frac{1}{1 + \exp(-t)}$$



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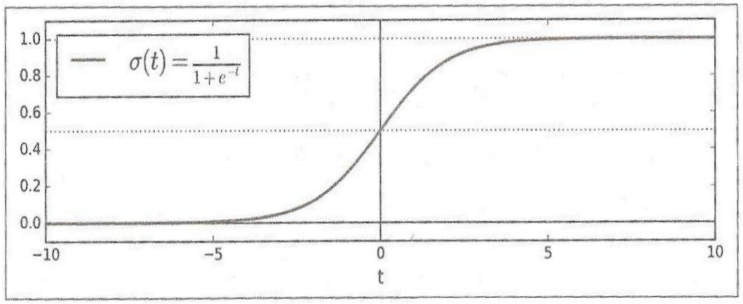
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Machine Learning – Logistic regression

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Logistic Regression model prediction

$$\hat{y} = \begin{cases} 0 \text{ if } \hat{p} < 0.5, \\ 1 \text{ if } \hat{p} \geq 0.5. \end{cases}$$

p. 143 in "Hands on ML" by Geron

Machine Learning – Logistic regression

Equation 4-16. Cost function of a single training instance

$$c(\theta) = \begin{cases} -\log\left(\hat{p}\right) & \text{if } y = 1, \\ -\log\left(1 - \hat{p}\right) & \text{if } y = 0. \end{cases}$$

Where:

Logistic Regression model prediction

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p. 144 in "Hands on ML" by Geron

Machine Learning – Logistic regression

Logistic regression

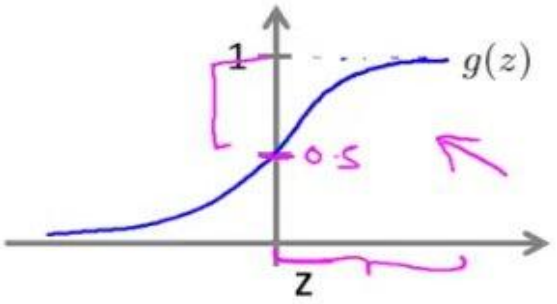
$h_\theta(x) = g(\theta^T x) = P(y=1 \mid x; \theta)$

$g(z) = \frac{1}{1+e^{-z}}$

Suppose predict "$y = 1$" if $h_\theta(x) \geq 0.5$

$\theta^T x \geq 0$

predict "$y = 0$" if $h_\theta(x) \leq 0.5$



$g(z) \geq 0.5$ when $z \geq 0$

$h_\theta(x) = g(\theta^T x) \geq 0.5$ whenever $\theta^T x \geq 0$

(z)

Andrew Ng. Coursera.

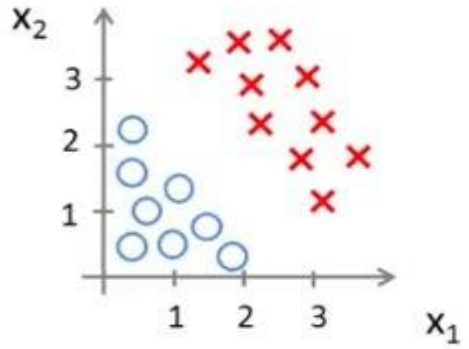
A little help from chatGPT.

Exercise:
Logistic Regression and chatGPT.

Try and see if chatGPT can help you understand what logistic Regression is…

Machine Learning – Logistic regression

Decision Boundary



$\theta = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}$

$$h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

$-3 \quad 1 \quad 1$

Predict "$y = 1$" if $-3 + x_1 + x_2 \geq 0$

$\theta^T x$

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Machine Learning – Logistic regression

Cost function:

Equation 4-17. Logistic Regression cost function (log loss)

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}log\left(\hat{p}^{(i)}\right) + \left(1 - y^{(i)}\right)log\left(1 - \hat{p}^{(i)}\right)\right]$$

The bad news is that there is no known closed-form equation to compute the value of θ that minimizes this cost function (there is no equivalent of the Normal Equation). But the good news is that this cost function is convex, so Gradient Descent (or any other optimization algorithm) is guaranteed to find the global minimum (if the learning rate is not too large and you wait long enough). The partial derivatives of the cost function with regards to the jth model parameter θ_j is given by Equation 4-18.

Equation 4-18. Logistic cost function partial derivatives

$$\frac{\partial}{\partial\theta_j}J(\theta) = \frac{1}{m}\sum_{i=1}^{m}\left(\sigma\left(\theta^T \cdot \mathbf{x}^{(i)}\right) - y^{(i)}\right)x_j^{(i)}$$

p. 145 in "Hands on ML" by Geron

Machine Learning – Logistic regression

Logistic regression cost function

$$J(\theta) = \frac{1}{m} \sum_{i=1}^{m} \text{Cost}(h_\theta(x^{(i)}), y^{(i)})$$

$$\text{Cost}(h_\theta(x), y) = \begin{cases} -\log(h_\theta(x)) & \text{if } y = 1 \\ -\log(1 - h_\theta(x)) & \text{if } y = 0 \end{cases}$$

Note: $y = 0$ or 1 always

$$\text{Cost}(h_\theta(x), y) = -y \log(h_\theta(x)) - (1-y) \log(1 - h_\theta(x))$$

If $y = 1$: $\text{Cost}(h_\theta(x), y) = -\log h_\theta(x)$

If $y =$

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Machine Learning – Logistic regression

Gradient Descent

$$J(\theta) = -\frac{1}{m}[\sum_{i=1}^{m} y^{(i)} \log h_\theta(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_\theta(x^{(i)}))]$$

Want $\min_\theta J(\theta)$:

Repeat {

$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta)$$

(simultaneously update all θ_j)

}

$$\frac{\partial}{\partial \theta_j} J(\theta) = \frac{1}{m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

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Machine Learning – Logistic regression

But don't worry :)

We will just use SciKit-Learn's build-in functions !

Machine Learning – Logistic regression

Exercises

3.1 & 3.2

Machine Learning – The Iris DataSet



https://archive.ics.uci.edu/ml/datasets/Iris

Machine Learning – The Iris DataSet

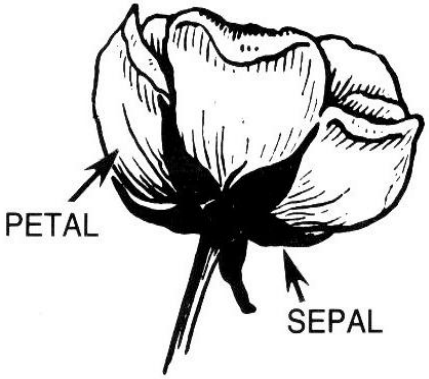
”This is perhaps the best known database to be found in the pattern recognition literature”
”A good project because it is so well understood”.

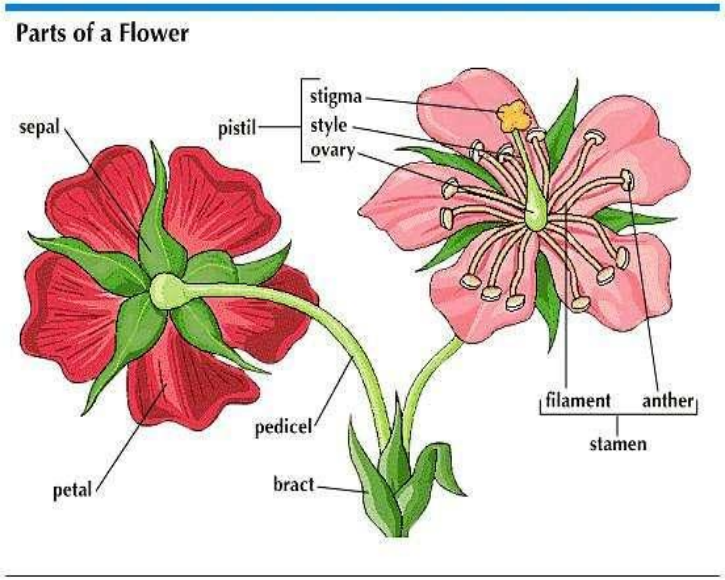
- The Iris Dataset is a classification problem, allowing you to practice
with perhaps an easier type of supervised learning algorithm.

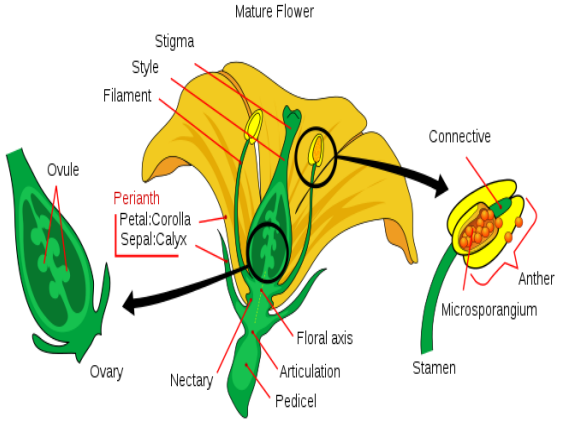
- It only has 4 attributes and 150 rows, meaning
it is small and easily fits into memory (and a screen or A4 page).

- All of the numeric attributes are in the same units and the same scale,
not requiring any special scaling or transforms to get started.

Machine Learning – The Iris DataSet







https://archive.ics.uci.edu/ml/datasets/Iris

Machine Learning – Support Vector Machines

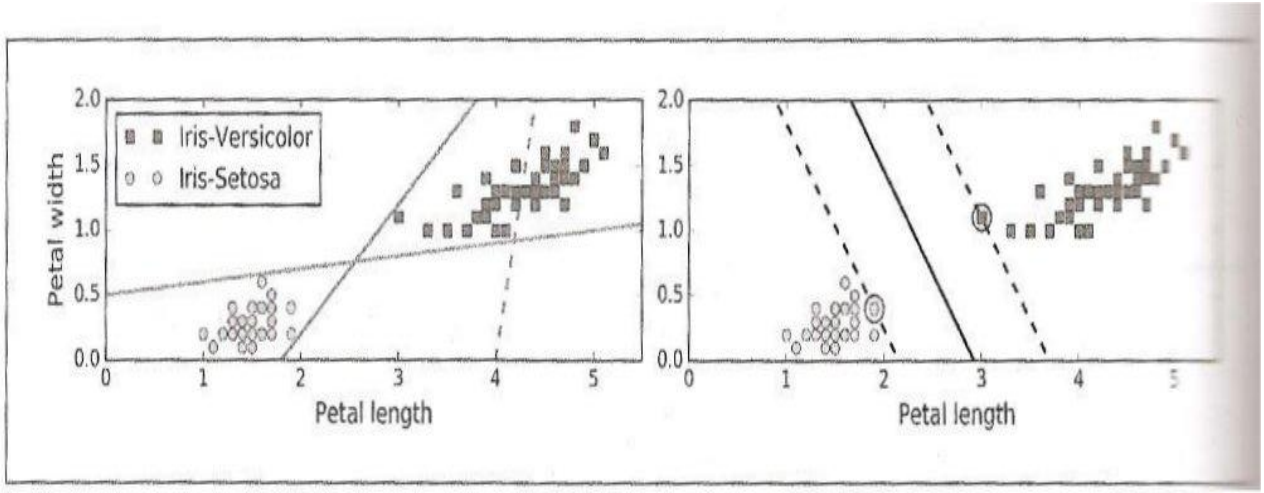


Figure 5-1. Large margin classification

You can think of an SVM classifier as fitting the widest possible street (between the classes). This is calles "large margin classification".

p. 156 in "Hands on" by Geron.

Machine Learning – Support Vector Machines

Use SVM software package (e.g. liblinear, libsvm, ...) to solve for parameters θ.

Need to specify:

→ Choice of parameter C.

Choice of kernel (similarity function):

E.g. No kernel ("linear kernel")

Predict "y = 1" if $\theta^T x \geq 0$

$\theta_0 + \theta_1 x_1 + \cdots + \theta_n x_n \geq 0$

→ n large, m small

$x \in \mathbb{R}^{n+1}$

Gaussian kernel:

$$f_i = \exp\left(-\frac{||x - l^{(i)}||^2}{2\sigma^2}\right), \text{ where } l^{(i)} = x^{(i)}.$$

Need to choose σ^2.

$x \in \mathbb{R}^n$, n small and/or m large



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Machine Learning – Support Vector Machines

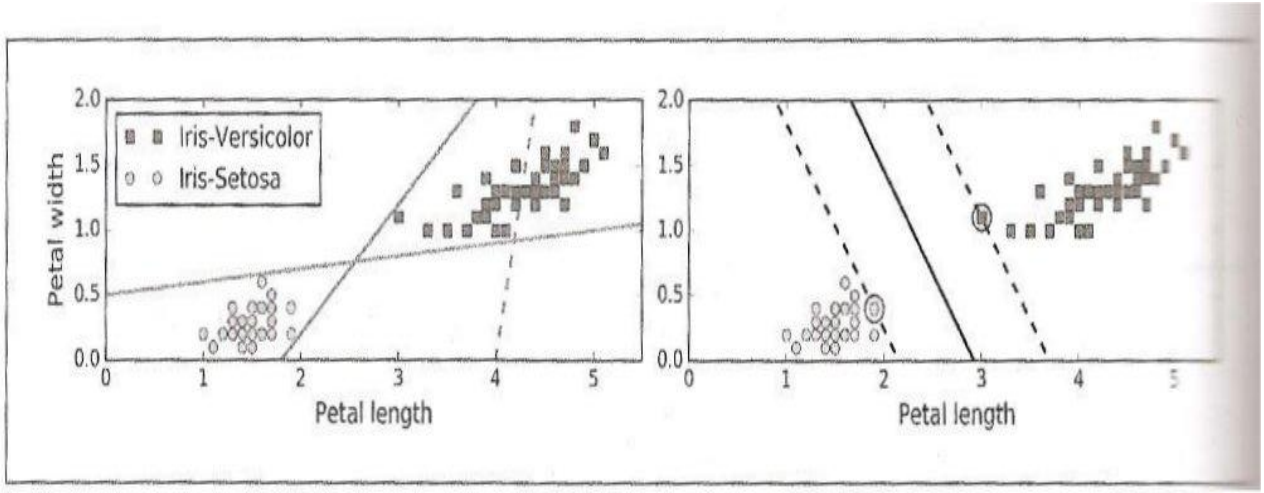


Figure 5-1. Large margin classification

Many datasets are not linearly separable.
SVMs can handle some cases of this by adding ”kernels”.
We will look at an example in exercise 2.6, but won't go into the math.

See p. 154 in "Hands on" by Geron.

Machine Learning – Logistic regression

Remaining Exercises...

Homework

Do the exercises for this week.

Read:
"Hands-On Machine Learning" by Aurelien Geron.
p. 164 - 170 (142-147 i v2) Logistic Regression.
p. 175 - 185 (153-161 i v2) Support Vector Machines.

"Real-World Machine Learning" by Brink et al.
Chapter 3. (What is Machine Learning, Real World Data).
p. 52 - 63
p. 64 – 68 (aside about SVMs)
p. 69 - 76
p. 78 – 84 Overfitting, underfitting, cross validation
(repeat from week 2)